

Systematic review and network meta-analysis of machine learning algorithms in sepsis prediction

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ABSTRACT

Background: With the integration of artificial intelligence and clinical medicine, machine learning (ML) algorithms have been applied to develop sepsis predictive models for sepsis management. The purpose is to systematically summarize existing evidence to determine the effectiveness of ML algorithms in sepsis.

Methods: We conducted a systematic electronic search of databases including PubMed, Cochrane Library, Embase, and the Web of Science, and included all case-control and cohort studies using terms reflecting sepsis and ML up to September 2023. statistical software STATA was used for network meta-analysis, and QUADAS-2 tool was used to assess the certainty of evidence.

Results: The SUCRA results for sensitivity, specificity, and predictive accuracy of various models are as follows: DSPA (77.0 %) > Imbalance-XGBoost (72.9 %) > CNN + Bi-LSTM (69.7 %) > CNN (67.3 %) > LR (62.4 %) > Ensemble model (55.9 %) > RF (53.2 %) > ET (51.3 %) > XGBoost (49.1 %) > DNN (48.1 %) > MLP (47.5 %) > RBF (47.1 %) > KNN (45.8 %) > NB (33.3 %) > SVM (13.7 %) > Bi-LSTM (5.7 %); CNN (78.3 %) > CNN + Bi-LSTM (77.6 %) > DSPA (75.1 %) > ET (69 %) > Bi-LSTM (68.5 %) > MLP (51 %) > RBF (50.2 %) > KNN (47.3 %) > RF (47 %) > Ensemble Model (43.4 %) > XGBoost (38.1 %) > SVM (37.3 %) > NB (34.2 %) > DNN (31.1 %) > LR (30.4 %) > Imbalance-XGBoost (21.5 %); DSPA (85.9 %) > CNN + Bi-LSTM (82.6 %) > CNN (81.9 %) > Imbalance-XGBoost (76.8 %) > ET (67.8 %) > RF (51.1 %) > Ensemble model (47.7 %) > XGBoost (44.4 %) > LR (42.7 %) > MLP (38.1 %) > RBF (37.8 %) > KNN (37.3 %) > DNN(35.8 %) > Bi-LSTM(33.3 %) > NB(21.5 %) > SVM(15.3 %).

Conclusions: DSPA and CNN may be the best ML algorithms for predicting sepsis. Imbalance-XGBoost algorithm outperformed other traditional ML algorithms in terms of sensitivity and predictive accuracy. This study has several implications for clinical practice and research, highlighting the potential benefits of using ML algorithms in sepsis management, particularly in improving sepsis detection and reducing mortality rates. Through our systematic review and network meta-analysis, we have provided a comprehensive and accurate assessment of the effectiveness of ML algorithms in sepsis prediction, emphasizing the need for further exploration and evaluation of these algorithms to advance sepsis management.

Abbreviations: AI, artificial intelligence; Bi-LSTM, bidirectional long- and short-term memory networks; CNN, convolutional neural networks; DL, deep learning; DNN, deep neural networks; DSPA, deep SOFA score -sepsis prediction algorithm; FN, false negative; FP, false positive; KNN, K-nearest neighbour; LR, logistic regression; ML, machine learning; MLP, Multilayer Perceptron; NB, Naive Bayes; NMA, network meta-analysis; RBF, Radial Basis Function; RFDT, Random Forest Decision Trees; RNN, recurrent neural networks; Se, sensitivity; SOFA, sequential organ failure assessment; Sp, specificity; SVM, support vector machines; TN, true negative; TP, true positive; XGB, Extreme Gradient Boosting.

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1. Introduction

Sepsis management remains a significant challenge for global healthcare systems (Evans et al., 2021). Although age-standardized morbidity and mortality rates have decreased, sepsis remains a leading cause of global health loss (Rudd et al., 2020). Early identification of sepsis in patients at high risk is crucial to improve their outcomes. Methods to improve early identification of sepsis may reduce the severity and economic burden on patients and their families (Komorowski et al., 2022; Yan et al., 2022).

Machine learning (ML) algorithms have the ability to learn from data and make predictions (An et al., 2023). They have been applied to improve healthcare quality and efficiency, including diagnosis, prognosis, and treatment (Krishnan et al., 2022; Ngiam & Khor, 2019). Compared to traditional methods such as clinical scores or biomarkers, ML algorithms can predict sepsis patients more accurately and efficiently (Eskandari et al., 2023; Zhang et al., 2023). ML algorithms include logistic regression (LR), support vector machines (SVM), decision trees, random forests (RF), neural networks, and deep learning (DL). They can use a wide range of data including demographic, clinical, laboratory, vital signs, imaging, and genomic data to learn sepsis-related patterns and features (Guo et al., 2022). This allows for a more comprehensive and accurate assessment of sepsis risk and prediction of patient outcomes.

Traditional *meta-analysis* is a systematic review that integrates the results of multiple studies to provide a comprehensive summary of the evidence. It provides an overall view of the research findings by combining the results of all the studies (Umscheid, 2013). On the other hand, network *meta-analysis* (NMA) extends traditional *meta-analysis* by incorporating the relationships between studies into the analysis. In

NMA, studies are represented as nodes in a network, and relationships between them are represented as edges (Doosti-Irani et al., 2021). This allows for a more comprehensive understanding of the research landscape and a better evaluation of the relative efficacy of different algorithms, which can help to identify the most effective approach for sepsis management.

Although ML has been widely adopted in sepsis management and has been validated using traditional *meta-analysis*, there are limitations in comprehensiveness and accuracy due to the heterogeneity of studies and data (Table 1). ML algorithms can be complex and difficult to interpret, and their results can be affected by data quality and model selection decisions (Ngiam & Khor, 2019; Riccardi et al., 2014). In addition, traditional *meta-analysis* may not be able to fully capture the complex relationships between studies (Hutton et al., 2015). On the contrary, NMA has the potential to overcome these limitations and provide a more accurate and comprehensive evaluation of ML algorithms in sepsis management. NMA have the ability to integrate multiple studies, thereby increasing sample size and reducing the risk of bias (Doosti-Irani et al., 2021). NMA allows comparative analyses of different ML algorithms, which can help to identify the most effective algorithms (Liu et al., 2023). In addition to traditional *meta-analysis* functionality, NMA can also be used to analyze rankings based on algorithmic efficacy, which is not possible with traditional *meta-analysis* (Hutton et al., 2015). NMA can better handle heterogeneity because it incorporates multiple studies, and it can more accurately estimate the efficacy of ML algorithms by covering different types of ML algorithms (Liu et al., 2023). Therefore, systematic reviews and NMA may provide a more comprehensive and accurate evaluation of ML algorithms by integrating multiple studies, analyzing their results simultaneously, and incorporating their relationships into the analysis.

Table 1

Summary of published systematic reviews and traditional *meta-analyses* of publications on “ML in sepsis management”.

Title	Year	Methods	Primary ML	Major conclusions	Compared to network <i>meta-analysis</i>	
					Advantages	Disadvantages
Machine Learning for the Prediction of Sepsis: A Systematic Review and Meta-Analysis of Diagnostic Test Accuracy (Fleuren et al., 2020)	2020	Systematic Review and Meta-Analysis	Supervised Learning, Unsupervised Learning, DL, etc.	ML models have high accuracy and reliability in predicting sepsis.	This approach provides a more comprehensive analysis of existing studies.	There may be differences in data acquisition and interpretation, and more research is needed to validate the findings and to understand the performance of the models in different datasets and the limitations of various ML algorithms.
Predicting Sepsis Onset in ICU Using Machine Learning Models: A Systematic Review and Meta-Analysis (Yang et al., 2023)	2023	Systematic Review and Meta-Analysis	DL (Neural Networks), etc., combined with various ML techniques	Through DL and <i>meta-analysis</i> , it is possible to accurately predict the onset of sepsis in ICU.		
Artificial intelligence in the management and treatment of burns: A systematic review and <i>meta-analyses</i> (Taib et al., 2023)	2023	Systematic Review and Meta-Analyses	DL, Unsupervised Learning, Supervised Learning, etc.	ML models have potential for improving burn management and treatment.		
Sepsis Prediction, Early Detection, and Identification Using Clinical Text for Machine Learning: A Systematic Review (Yan et al., 2022)	2022	Systematic Review	Various ML Algorithms (e.g., Supervised Learning, Unsupervised Learning, DL)	ML algorithms have potential for predicting, early detection, and identifying sepsis using clinical text.	A more comprehensive assessment of existing research, including all relevant studies, is provided, as well as an evaluation of the performance of different algorithms.	More research is needed to evaluate the generalisation ability and stability of ML algorithms (especially deep ML).
Early Prediction of Sepsis in the ICU Using Machine Learning: A Systematic Review (Moor et al., 2021)	2021	Systematic Review	Various ML Algorithms (e.g., Supervised Learning, Unsupervised Learning)	ML algorithms have potential for early prediction of sepsis in the ICU.		
Prediction of Sepsis Patients Using Machine Learning Approach: A Meta-Analysis (Islam et al., 2019)	2019	Meta-Analysis	DL (Neural Networks), etc.	Through ML, it is possible to predict sepsis patients with a certain degree of accuracy.		

Sensitivity (Se), specificity (Sp), and predictive accuracy are essential indicators for assessing the accuracy of ML algorithms in the early identification of sepsis (Bunn et al., 2021; Eskandari et al., 2023; Huang et al., 2021; Liu et al., 2022; Tran et al., 2020; Zhang et al., 2023). These indicators enable a comparison of different algorithms and inform efforts to improve the early identification of sepsis, thereby reducing its incidence and mortality rates.

Therefore, further exploring and assessing the efficacy of ML algorithms in sepsis prediction using NMA is crucial. This paper will comprehensively assess the efficacy of ML algorithms in sepsis prediction through a systematic review and NMA to clarify the application of ML in sepsis management and identify the most effective algorithms. This approach will provide a more comprehensive and accurate evaluation of ML algorithms, enabling the identification of the most reliable and effective algorithms for sepsis prediction. We found that ML algorithms have varying degrees of effectiveness in predicting sepsis. Among the identified algorithms, some performed well in predicting sepsis, indicating that further exploration and evaluation of these algorithms may improve sepsis management and reduce mortality.

2. Materials and methods

The study involved a systematic review and NMA of ML algorithms in sepsis prediction, using a specially designed data collection tool in Microsoft Office Excel software to extract data independently. Information was collected, and any missing data were resolved by contacting the authors for further information. Subsequently, the data were entered into Stata 17.0 software for analysis.

2.1. Design

We registered our protocol detailing the strategy used to search the literature. This systematic review and NMA protocol was prospectively registered in the international prospective register of systematic reviews of the University of York (PROSPERO, Registration No: CRD42023466052, **Additional file 1**) and was written following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA, **Additional file 2**) extension statement for reporting systematic review incorporating NMA (Hutton et al., 2015).

2.2. Search strategy

This systematic review and NMA comprehensively searched multiple databases and reviewed expert opinions to identify relevant trials on sepsis-related ML algorithms. Two review authors (YLG and YCL) searched PubMed, The Cochrane Library, Embase, and Web of Science from inception to September 2023, according to the search rules of each database, using terms reflecting sepsis and ML. In addition, they reviewed expert opinions, references of selected studies, and previous systematic reviews. Other potentially eligible trials were identified by searching the reference lists of included trials, relevant systematic reviews, meta-analyses, and technology assessment reports. The searched keywords mainly include “Sepsis”, “Septic”, “Bloodstream Infection”, “Infection, Bloodstream”, “Pyemia”, “Pyohemia”, “Pyemia”, “Septicemia”, “Poisoning, Blood”, “Blood Poisoning”, “Severe Sepsis”, “Machine Learning”, “Learning, Machine”, “Transfer Learning” and “Learning, Transfer”. The PubMed Medline filter search filters were applied to the databases. The detailed search strategy was as follows: (“Sepsis”[Mesh]) OR (((((((Bloodstream Infection[Title/Abstract]) OR (Infection, Bloodstream[Title/Abstract])) OR (Pyemia[Title/Abstract])) OR (Pyohemia[Title/Abstract])) OR (Pyemia[Title/Abstract])) OR (Septicemia[Title/Abstract])) OR (Poisoning, Blood[Title/Abstract])) OR (Blood Poisoning[Title/Abstract])) OR (Severe Sepsis [Title/Abstract])) AND (“Machine Learning”[Mesh]) OR (((Machine Learning[Title/Abstract]) OR (Learning, Machine[Title/Abstract])) OR (Transfer Learning[Title/Abstract])) OR (Learning, Transfer[Title/

Abstract]))).

2.3. Inclusion/exclusion criteria

The following NMA will focus on the evaluation of ML algorithms used to predict sepsis using original research studies that met the inclusion criteria. The literature identified through the four databases was combined, duplicates removed, and the remaining articles were original research (excluding letters, editorials, research protocols, case reports, non-clinical studies, meta-analyses, etc.) and written in English. The study designs were either case-control or cohort studies with control groups. The PICO criteria (population, intervention, comparison, and outcome) were applied to all studies to determine their eligibility for inclusion. Inclusion criteria were: 1. Patients in the literature had no limitations regarding age, gender, race, or duration of illness; 2. The study intervention involved using a different machine model to predict the occurrence of sepsis; 3. The patients in the literature had a diagnosis of sepsis that was compatible with the current criteria (Bone et al., 1992; Evans et al., 2021; Levy et al., 2003; Singer et al., 2016) or as defined by the study authors; and 4. The study outcome metrics mainly consisted of Se, Sp, true positive (TP), true negative (TN), predictive accuracy (TP + TN), false positive (FP), false negative (FN), and sample size. Wherefore, where TP, TN, FP, or FN were not reported, they were calculated from known variables (Se and Sp).

2.4. Study selection

ML algorithms are one of the most important components of artificial intelligence (AI) technology. The studies that met the inclusion criteria were reviewed and evaluated using different ML algorithms to construct predictive or diagnostic models for sepsis patients. 16 different ML algorithms were included, including traditional ML algorithms such as LR, SVM, and RF, and DL algorithms such as convolutional neural networks (CNN) and bidirectional long- and short-term memory networks (Bi-LSTM). The literature retrievals were imported into Endnote software (Thomson Research Soft) and duplicates were removed. Two review authors (YLG and CLW) independently assessed whether the titles, abstracts, and full texts of retrieved literature met the inclusion/exclusion criteria formulated above by first reading titles and abstracts and then reading full texts. The full text of all potentially relevant study reports was retrieved for further assessment of eligibility. Disagreements between reviewers were first resolved by consensus conference, and if consensus could not be reached, a third reviewer (YFC) was the arbitrator. Finally, two review authors (CLW and ZYW) carefully read the full texts in detail and removed literature that did not report outcome measures, had incomplete data, or had duplicate data results.

2.5. Data extraction

To ensure the accuracy and rigor of the research, data were extracted independently by three review authors (CLW, ZYW, and JXS) from included studies using a specially designed data collection tool in Microsoft Office Excel software. Information such as author, year of literature, study design, title, journal name, country, inclusion criteria, type of ML algorithm, Se, Sp, TP, TN, FP, FN, and sample size were collected. If any missing data were encountered, attempts were made to contact the authors for further information. Any disagreement was discussed with two other review authors (YFC and YLG) according to the screening criteria. The data were then entered into Stata 17.0 software by two review authors (CLW and JXS), and the data were reviewed by two other review authors (YFC and YLG) to ensure accuracy.

2.6. Literature quality assessment

The purpose of this step was to ensure the accuracy and rigor of the review by assessing the quality of the included studies and their

applicability to the review question. Two review authors (YLG and ZYW) independently used the Quality Assessment of Diagnostic Accuracy Studies (QUADAS)-2 tool (Whiting et al., 2011) to perform an initial assessment of the risk of bias and clinical applicability of the studies. After team discussion and consensus, they completed the assessment. The QUADAS-2 form was categorized into four components: “patient selection”, “trials to be evaluated”, “gold standard” and “case flow and progress”. The risk of bias was assessed for all components, and the first three components had an assessment of clinical applicability. The risk of bias was considered to be “low”, “high” or “uncertain” based on 14 entries (3 non-essential entries suggested by Cochrane were removed from this study).

2.7. Data analysis

The NMA was conducted using Stata 17.0 software. The measurements were expressed as the relative risk (RR) and its 95 % confidence interval (CI). The overall consistency between the direct and indirect evidence was analyzed using the inconsistency test; $P > 0.05$ was considered to indicate that there was no consistency, and a consistency model was fitted, and vice versa for the inconsistency model. Local inconsistency tests for direct and indirect comparisons were performed using node splitting, and $P < 0.05$ was considered present. The efficacy of the intervention was ranked according to the cumulative probability in terms of the area under the subsurface cumulative ranking (SUCRA), with a larger area under the curve indicating a better accuracy of the model in predicting sepsis. To assess potential publication bias and small sample effects among included studies, funnel plots were used.

3. Result

3.1. Study characteristics

The flowchart of the study is shown in Fig. 1. Following the search strategy, a total of 1492 records were retrieved for this study, and 359 records were obtained after 1133 duplicates were removed by Endnote X9 software. After removing 334 irrelevant studies, uncontrolled

experimental studies, conferences, abstracts, meta-analyses, and other literature based on reading the title and abstract, a total of 25 records remained. Then, for the remaining literature, after removing 18 records for non-compliance with outcome indicators by reading the full text, a total of 7 records were included. All of the studies were cohort studies. The details of the characteristics of the included studies are given in Table 2. The sample sizes of the studies ranged from 211 to 22,314. This NMA included 16 ML algorithms, including “Bi-LSTM”, “CNN”, “CNN + Bi-LSTM”, “Deep Neural Networks (DNN)”, “Deep Sequential Organ Failure Assessment (SOFA) score -Sepsis Prediction Algorithm (DSPA)”, “LR”, “Ensemble Model [(LR, RF Decision Trees (RFDT), and Extreme Gradient Boosting (XGB))]”, “Extremely Randomized Trees (ET)”, “Gradient Boosting Machine (XGBoost)”, “Imbalance-XGBoost”, “K-Nearest Neighbor (KNN)”, “Multilayer Perceptron (MLP)”, “Naive Bayes (NB)”, “Radial Basis Function (RBF)”, “RF”, and “SVM”.

3.2. Quality assessment of the inclusion studies

The risk of bias and the clinical applicability of the included studies are summarized in Table 3. The overall quality of this systematic review and NMA is satisfactory. All of the studies were considered to have a high risk of bias in at least one of the domains (Whiting et al., 2011). There were domains of the studies in which the risk of bias and the clinical applicability were not clear because they were not mentioned in the text or because they were described poorly.

Q1: whether consecutive or randomized cases were included; Q2: whether case-control study design was avoided; Q3: whether inappropriate exclusions were avoided; Q4: whether the interpretation of results was confounded by the results of the gold standard; Q5: whether thresholds were set a priori; Q6: whether the gold standard correctly differentiated between disease states; Q7: whether blinding was used in the interpretation of the gold standard; Q8: whether there was an appropriate time interval between the trial to be evaluated and the gold standard; Whether the gold standard correctly discriminated between disease states; Q9 to Q10: Whether all patients received the same type of gold standard; and Q11: Whether all cases were included in the analysis. “√”, “x”, and “?” are considered “low”, “high”, and “uncertain” risks,

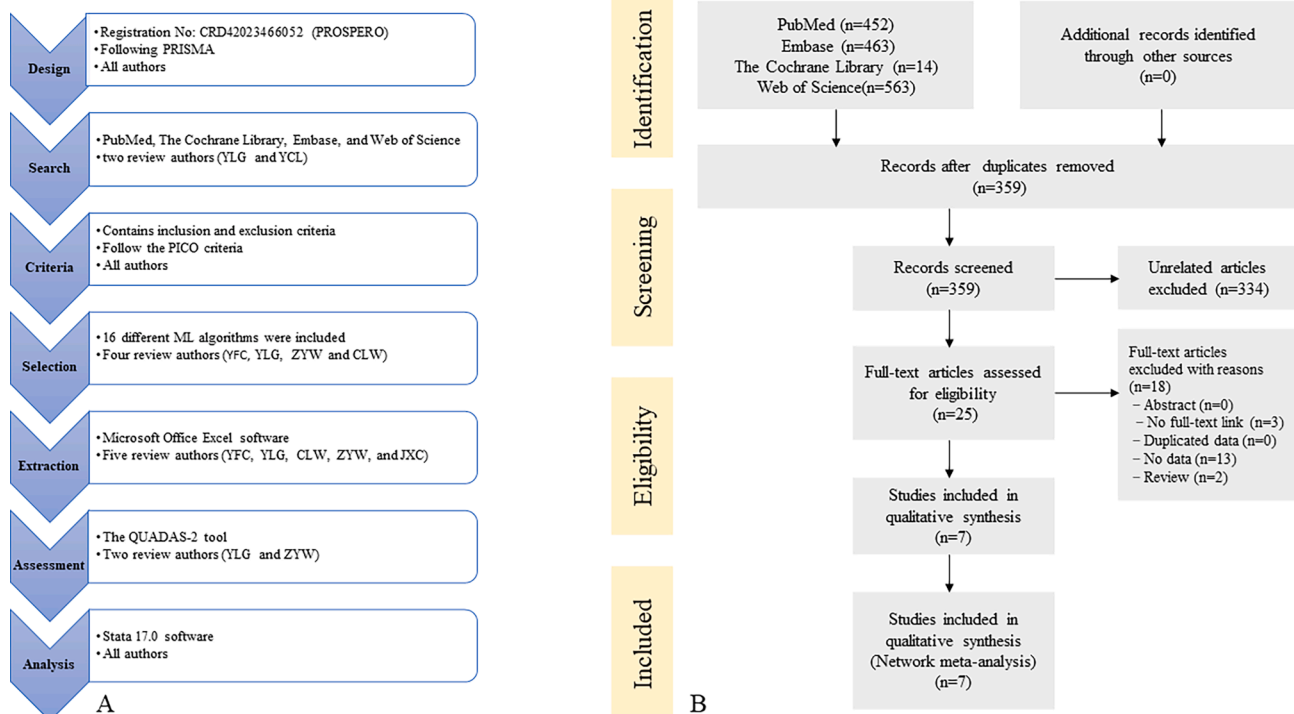


Fig. 1. Technology roadmap for research methodologies (A). PRISMA flowchart diagram of study selection (B, undertaken in September 2023).

Table 2
Basic characteristics of the inclusion studies.

Rank	Literature	Year	Study design	Country	Sample size		Criteria	ML algorithm	Se	Sp	Sample size					Total sample size
					Sepsis	Without sepsis					Tp	Tn	Tp + Tn	Fp	Fn	
1	(Asuroglu & Ogul, 2021)	2020	Cohort study	Turkey	1,404	3,750	☆	CNN	0.873	0.981	1,226	3,679	4,904	71	178	5,154
								RF	0.943	0.935	1,324	3,506	4,830	244	80	
								DSPA	0.984	0.968	1,382	3,630	5,012	120	22	
								Bi-LSTM	0.402	0.954	564	3,578	4,142	173	840	
								CNN + Bi-LSTM	0.897	0.978	1,259	3,668	4,927	83	145	
								SVM	0.587	0.728	824	2,730	3,554	1,020	580	
								LR	0.737	0.863	1,035	3,236	4,271	514	369	
								KNN	0.614	0.955	862	3,581	4,443	169	542	
2	(Zhang et al., 2023)	2023	Cohort study	China	550	568	△	NB	0.607	0.793	852	2,974	3,826	776	552	1,118
								RF	0.71	0.68	391	386	777	182	160	
								XGBoost	0.7	0.63	385	358	743	210	165	
								ET	0.68	0.71	374	403	777	165	176	
								Imbalance-XGBoost	0.86	0.59	473	335	808	233	77	
								SVM	0.46	0.62	253	352	605	216	297	
								KNN	0.9807	1.00	19,779	20,168	39,947	0	389	
3	(Eskandari et al., 2023)	2023	Cohort study	Iran	20,168	20,168	☆									40,336
4	(Bunn et al., 2021)	2020	Cohort study	USA	2,141	221,073	▽	RBF	0.9677	1.00	19,517	20,168	39,685	0	651	223,214
								MLP	0.9687	1.00	19,537	20,168	39,705	0	631	
								LR	0.6261	0.6509	1,340	143,896	145,237	77,177	801	
								SVM	0.5142	NA	1101	NA	NA	NA	1,040	
								RF	0.6953	0.6067	1,489	134,125	135,614	86,948	652	
								XGB	0.7030	0.6667	1,505	147,389	148,894	73,684	636	
								Ensemble model (LR, RFD, XGB)	0.7074	0.6454	1,515	142,681	144,195	78,392	626	
5	(Huang et al., 2021)	2020	Cohort study	USA	868	2,032	◇	LR	0.62	0.73	538	1,483	2,022	549	330	2,900
6	(Ibrahim et al., 2020)	2020	Cohort study	UK	4,256	9,472	☆	KNN	0.45	0.73	391	1,483	1,874	549	477	13,728
								RF	0.38	0.88	330	1,788	2,118	244	538	
								XGBoost	0.4	0.85	347	1,727	2,074	305	521	
								SVM	0.13	0.95	113	1,930	2,043	102	755	
								RF	0.85	0.79	3,618	74,83	11,100	19,89	638	
7	(Tran et al., 2020)	2020	Cohort study	USA	92	KNN	0.96	XGBoost	0.87	0.81	3,703	7,672	11,375	1,800	553	
								SVM	0.86	0.83	3,660	7,862	11,522	1,610	596	
								0.88	88	105	193	14	4	211		
								LR	0.98	0.83	90	99	189	20	2	
								NB	0.94	0.87	86	104	190	15	6	
								RF	0.96	0.81	88	96	185	23	4	
								DNN	1.00	0.80	92	95	187	24	0	
								SVM	0.98	0.82	90	98	188	21	2	
								XGBoost	0.96	0.76	88	90	179	29	4	

Notes:

☆ Sepsis-3 (Singer et al., 2016). defines the criteria for severe infection and organ failure.

△ Patients with positive results of body fluid culture and antibiotics are considered infected. Infected patients with Sequential Organ Failure Assessment (SOFA) score ≥ 2 are considered sepsis patients (Zhang et al., 2023).

▽ Specifically, for the diagnosis of sepsis, either preoperative or postoperative, the American College of Surgeons National Surgical Quality Improvement Program (ACS-NSQIP) requires the presence of documented infection or end-organ ischemia in addition to at least 2 of the 5 clinical signs and symptoms of systemic inflammatory response (Bunn et al., 2021).

◇ Patients were classified as positive for sepsis if either (a) the patient was culture positive for a bacterial pathogen within 5 days of being evaluated for sepsis (either from blood, urine, cerebrospinal fluid, peritoneum, or pleural fluid) and was treated accordingly with a minimum of 5 days (or 120 h) of antibiotic treatment or (b) the patient had no positive culture but still was treated as clinically positive, having received a minimum of 120 h of antibiotics (Huang et al., 2021).

Table 3

Summary of risk of bias and clinical applicability of the seven included studies.

Literature	Patient selection			Trials to be evaluated		Gold standard			Case flow and progress		
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
(Asuroglu & Ogul, 2021)	?	✓	✓	✓	?	✓	✓	✓	✓	✓	×
(Zhang et al., 2023)	✓	✓	✓	?	?	✓	?	✓	✓	✓	×
(Eskandari et al., 2023)	?	✓	×	✓	?	✓	✓	✓	✓	✓	×
(Bunn et al., 2021)	✓	✓	?	✓	?	✓	✓	×	✓	✓	✓
(Huang et al., 2021)	×	✓	?	×	✓	✓	✓	×	✓	✓	✓
(Ibrahim et al., 2020)	×	✓	✓	?	?	✓	?	?	✓	✓	×
(Tran et al., 2020)	×	✓	?	?	✓	✓	?	?	×	×	×

Notes (Whiting et al., 2011):

respectively.

3.3. Network meta-analysis

3.3.1. Network map and inconsistency testing

To evaluate the predictive accuracy of an algorithm, Se and Sp are often combined in ML (Bunn et al., 2021; Tran et al., 2020). As shown in Fig. 2, the network evidence plots for Se, Sp, and predictive accuracy simultaneously indicated, directly or indirectly, that different ML algorithms were able to meet the basic conditions for achieving NMA. The results of the global inconsistency test (Additional file 3) for Se, Sp and predictive accuracy of the ML algorithm for sepsis prediction showed that the inconsistency test was not significant, and the difference was not statistically significant ($P > 0.05$). Furthermore, the local inconsistency test using the node-splitting method (Additional file 4) showed that the direct comparison results of the Se, Sp, and predictive accuracy were not inconsistent with the indirect comparison results, and the difference was not statistically significant ($P > 0.05$). Thus, consistency modeling analysis can be used for NMA. Different ML algorithms can meet the basic conditions for achieving NMA, and that the predictive accuracy of these algorithms is satisfactory.

3.3.2. Comparison of predictive value between different ML algorithms

The NMA results are reliable and can be used to compare the predictive accuracy of different ML algorithms. As shown in Table 4, the NMA results indicated that Bi-LSTM predicted sepsis with lower Se than CNN, DSPA, CNN + Bi-LSTM, Imbalance-XGBoost, LR, and RF, and the difference was statistically significant ($P < 0.05$). SVM predicted sepsis with lower Se compared to XGBoost, LR, and RF and the difference was statistically significant ($P < 0.05$). The remaining results were not statistically significant ($P > 0.05$) when comparing the Se of any two ML algorithms. As shown in Table 5, the NMA results showed that the Sp comparison between any two ML algorithms was not statistically significant ($P > 0.05$). As shown in Table 6, the NMA results indicated that SVM was less accurate in predicting sepsis than CNN + Bi-LSTM, CNN, DSPA, RF, and Imbalance-XGBoost, with a statistically significant difference ($P < 0.05$), and NB was less predictive accuracy in predicting sepsis than DSPA, with a statistically significant difference ($P < 0.05$). There was no statistically significant difference ($P > 0.05$) in the remaining predictive accuracy comparisons between any two machine models.

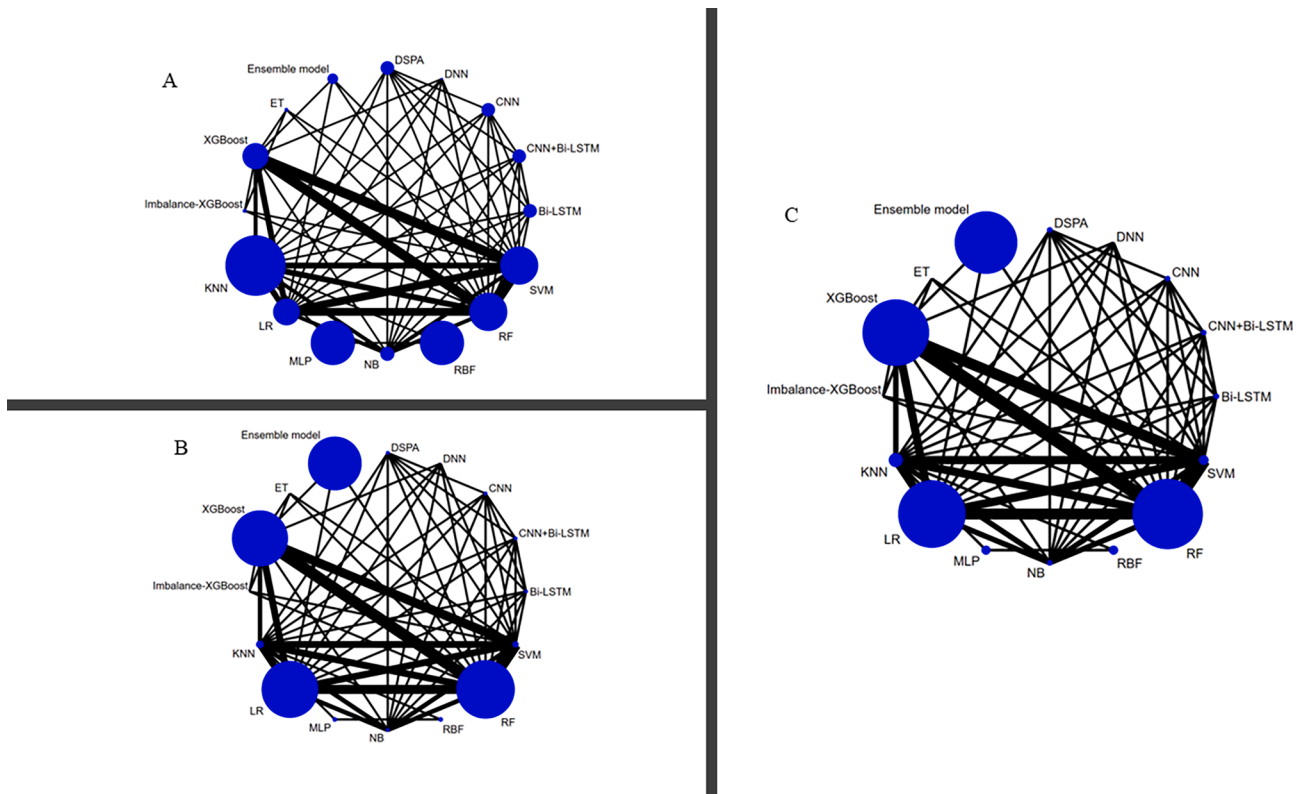
**Fig. 2.** Network plots of ML algorithms in sepsis management for Se (A), Sp (B), and predictive accuracy (C).

Table 4
Comparison of sensitivity between different ML algorithms.

Bi-LSTM															
0.45 (0.22,0.91)	CNN + Bi-LSTM														
0.46 (0.23,0.94)	1.03 (0.51,2.08)	CNN													
0.56 (0.26,1.20)	1.24 (0.57,2.68)	1.21 (0.56,2.61)	DNN												
0.41 (0.20,0.83)	0.91 (0.45,1.85)	0.89 (0.44,1.80)	0.73 (0.34,1.59)	DSPA											
0.51 (0.23,1.13)	1.14 (0.52,2.51)	1.11 (0.50,2.44)	0.92 (0.42,2.01)	1.25 (0.57,2.76)	Ensemble model										
0.54 (0.24,1.23)	1.21 (0.54,2.72)	1.18 (0.52,2.65)	0.97 (0.44,2.17)	1.33 (0.59,2.99)	1.06 (0.47,2.39)	ET									
0.55 (0.30,1.01)	1.21 (0.66,2.24)	1.18 (0.64,2.18)	0.98 (0.54,1.76)	1.33 (0.72,2.46)	1.07 (0.59,1.93)	1.00 (0.55,1.84)	XGBoost								
0.43 (0.19,0.97)	0.96 (0.42,2.15)	0.93 (0.41,2.10)	0.77 (0.35,1.72)	1.05 (0.47,2.36)	0.84 (0.37,1.89)	0.79 (0.39,1.61)	0.79 (0.43,1.44)	Imbalance-XGBoost							
0.56 (0.30,1.03)	1.24 (0.68,2.28)	1.21 (0.66,2.22)	1.00 (0.55,1.83)	1.36 (0.74,2.50)	1.09 (0.57,2.09)	1.03 (0.53,2.01)	1.02 (0.69,1.52)	1.30 (0.67,2.54)	KNN						
0.49 (0.27,0.90)	1.10 (0.61,1.99)	1.07 (0.59,1.94)	0.89 (0.49,1.60)	1.21 (0.67,2.19)	0.97 (0.53,1.76)	0.91 (0.47,1.74)	0.91 (0.63,1.29)	1.15 (0.60,2.20)	0.88 (0.60,1.31)	LR					
0.57 (0.22,1.44)	1.26 (0.50,3.20)	1.23 (0.48,3.11)	1.02 (0.40,2.57)	1.38 (0.55,3.51)	1.11 (0.43,2.89)	1.04 (0.39,2.76)	1.04 (0.46,2.33)	1.32 (0.50,3.49)	1.01 (0.50,2.05)	1.15 (0.51,2.57)	MLP				
0.63 (0.33,1.18)	1.40 (0.74,2.63)	1.36 (0.73,2.56)	1.13 (0.60,2.12)	1.54 (0.82,2.89)	1.23 (0.62,2.45)	1.16 (0.57,2.36)	1.15 (0.73,1.83)	1.46 (0.72,2.98)	1.13 (0.71,1.80)	1.27 (0.81,2.00)	1.11 (0.48,2.59)	NB			
0.57 (0.22,1.44)	1.26 (0.50,3.20)	1.23 (0.48,3.11)	1.02 (0.40,2.57)	1.38 (0.55,3.51)	1.11 (0.43,2.89)	1.04 (0.39,2.77)	1.04 (0.46,2.33)	1.32 (0.50,3.49)	1.01 (0.50,2.06)	1.15 (0.51,2.57)	1.00 (0.49,2.03)	0.90 (0.39,2.10)	RBF		
0.53 (0.29,0.95)	1.18 (0.66,2.12)	1.15 (0.64,2.06)	0.95 (0.53,1.69)	1.29 (0.72,2.32)	1.04 (0.58,1.87)	0.98 (0.53,1.79)	0.97 (0.71,1.32)	1.23 (0.67,2.25)	0.95 (0.65,1.38)	1.07 (0.77,1.50)	0.94 (0.42,2.08)	0.84 (0.54,1.30)	0.93 (0.42,2.08)	RF	
0.76 (0.42,1.37)	1.69 (0.94,3.04)	1.65 (0.92,2.96)	1.37 (0.77,2.43)	1.86 (1.03,3.34)	1.49 (0.83,2.68)	1.40 (0.76,2.57)	1.39 (1.02,1.90)	1.77 (0.97,3.24)	1.36 (0.94,1.98)	1.54 (1.10,2.16)	1.34 (0.60,2.99)	1.21 (0.78,1.87)	1.34 (0.60,2.98)	1.44 (1.07,1.92)	SVM

Table 5
Comparison of specificity between different ML algorithms.

Bi-LSTM																	
0.96 (0.75,1.23)	CNN + Bi-LSTM																
0.96 (0.75,1.23)	1.00 (0.77,1.29)	CNN															
1.15 (0.86,1.52)	1.19 (0.88,1.61)	1.20 (0.89,1.61)	DNN														
0.97 (0.76,1.25)	1.01 (0.78,1.31)	1.01 (0.78,1.31)	0.85 (0.63,1.14)	DSPA													
1.10 (0.82,1.46)	1.14 (0.85,1.54)	1.14 (0.85,1.54)	0.96 (0.70,1.30)	1.13 (0.84,1.52)	Ensemble model												
0.99 (0.74,1.33)	1.03 (0.76,1.40)	1.04 (0.76,1.41)	0.87 (0.63,1.19)	1.02 (0.75,1.39)	0.91 (0.67,1.24)	ET											
1.10 (0.89,1.36)	1.15 (0.92,1.44)	1.15 (0.92,1.45)	0.96 (0.76,1.22)	1.14 (0.91,1.43)	1.01 (0.80,1.26)	1.11 (0.88,1.40)	XGBoost										
1.19 (0.89,1.61)	1.24 (0.91,1.69)	1.25 (0.92,1.70)	1.04 (0.76,1.44)	1.23 (0.90,1.67)	1.09 (0.80,1.49)	1.20 (0.92,1.58)	1.08 (0.85,1.37)	Imbalance-XGBoost									
1.08 (0.88,1.31)	1.12 (0.90,1.40)	1.12 (0.90,1.40)	0.94 (0.74,1.20)	1.11 (0.89,1.38)	0.98 (0.77,1.26)	1.08 (0.84,1.40)	0.98 (0.84,1.13)	0.90 (0.70,1.17)	KNN								
1.13 (0.92,1.38)	1.17 (0.94,1.46)	1.18 (0.94,1.47)	0.98 (0.78,1.25)	1.16 (0.93,1.45)	1.03 (0.82,1.29)	1.14 (0.89,1.45)	1.02 (0.89,1.17)	0.94 (0.73,1.21)	1.05 (0.91,1.21)	LR							
1.06 (0.79,1.42)	1.10 (0.80,1.53)	1.11 (0.80,1.53)	0.93 (0.66,1.30)	1.09 (0.79,1.51)	0.97 (0.69,1.37)	1.07 (0.75,1.52)	0.96 (0.72,1.28)	0.89 (0.62,1.26)	0.98 (0.77,1.26)	0.94 (0.71,1.25)	MLP						
1.12 (0.90,1.39)	1.16 (0.92,1.47)	1.17 (0.92,1.47)	0.98 (0.76,1.26)	1.15 (0.91,1.45)	1.02 (0.78,1.33)	1.13 (0.86,1.47)	1.01 (0.85,1.21)	0.94 (0.71,1.23)	1.04 (0.87,1.24)	0.99 (0.84,1.18)	1.05 (0.78,1.42)	NB					
1.06 (0.79,1.42)	1.10 (0.80,1.53)	1.11 (0.80,1.53)	0.93 (0.66,1.30)	1.09 (0.79,1.51)	0.97 (0.69,1.37)	1.07 (0.75,1.52)	0.96 (0.72,1.28)	0.89 (0.62,1.26)	0.98 (0.77,1.26)	0.94 (0.71,1.25)	1.00 (0.77,1.30)	0.95 (0.70,1.28)	RBF				
1.08 (0.88,1.32)	1.12 (0.91,1.39)	1.13 (0.91,1.40)	0.94 (0.75,1.19)	1.11 (0.90,1.38)	0.99 (0.79,1.23)	1.09 (0.86,1.37)	0.98 (0.87,1.10)	0.90 (0.71,1.14)	1.00 (0.87,1.15)	0.96 (0.84,1.09)	1.02 (0.77,1.35)	0.97 (0.82,1.14)	1.02 (0.77,1.35)	RF			
1.11 (0.90,1.35)	1.15 (0.93,1.43)	1.15 (0.93,1.44)	0.97 (0.76,1.22)	1.14 (0.92,1.42)	1.01 (0.80,1.28)	1.11 (0.88,1.40)	1.00 (0.89,1.13)	0.93 (0.73,1.17)	1.03 (0.89,1.18)	0.98 (0.86,1.12)	1.04 (0.79,1.38)	0.99 (0.84,1.17)	1.04 (0.79,1.38)	1.02 (0.91,1.15)	SVM		

Table 6
Comparison of predictive accuracy between different ML algorithms.

Bi-LSTM																
0.84 (0.68,1.04)	CNN + Bi-LSTM															
0.84 (0.68,1.04)	1.00	CNN														
0.99 (0.79,1.25)	1.18	1.18	DNN													
0.83 (0.67,1.02)	0.98	0.98	0.83	DSPA												
0.96 (0.75,1.21)	1.14	1.13	0.96	1.16	Ensemble model											
0.90 (0.70,1.14)	1.07	1.06	0.90	1.08	0.94	ET										
0.97 (0.80,1.16)	1.15	1.14	0.97	1.17	1.01	1.08	XGBoost									
0.86 (0.67,1.10)	1.02	1.02	0.87	1.04	0.90	0.96	0.89	Imbalance-XGBoost								
0.98 (0.82,1.18)	1.17	1.16	0.99	1.19	1.03	1.10	1.02	1.14	KNN							
0.97 (0.81,1.16)	1.15	1.15	0.98	1.17	1.02	1.08	1.00	1.13	0.99	LR						
0.99 (0.75,1.30)	1.18	1.17	1.00	1.20	1.04	1.11	1.02	1.15	1.01	1.02	MLP					
1.03 (0.85,1.24)	1.23	1.22	1.04	1.25	1.08	1.15	1.07	1.20	1.05	1.06	1.04	NB				
0.99 (0.75,1.31)	1.18	1.17	1.00	1.20	1.04	1.11	1.02	1.15	1.01	1.02	1.00	0.96	RBF			
0.95 (0.80,1.13)	1.13	1.13	0.96	1.15	1.00	1.06	0.98	1.10	0.97	0.98	0.96	0.92	0.96	RF		
1.04 (0.88,1.25)	1.24	1.24	1.05	1.26	1.09	1.17	1.08	1.21	1.06	1.08	1.06	1.01	1.05	1.10	SVM	
	(1.04,1.48)	(1.04,1.47)	(0.88,1.26)	(1.06,1.51)	(0.90,1.32)	(0.97,1.41)	(0.98,1.19)	(1.01,1.46)	(0.95,1.19)	(0.97,1.20)	(0.83,1.34)	(0.89,1.16)	(0.83,1.34)	(1.00,1.21)		

3.3.3. Comparison of the probability of predicting value between different ML algorithms

The comparison of different ML algorithms based on their SUCRA scores is a useful tool for evaluating their performance in sepsis prediction. As shown in Table 7, probability ranking was used to compare the performance of different algorithms. SUCRA scores were used to indicate the average ranking of each algorithm, with higher SUCRA scores indicating a better-performing algorithm. The SUCRA results for Se showed that DSPA (77.0 %) > Imbalance-XGBoost (72.9 %) > CNN + Bi-LSTM (69.7 %) > CNN (67.3 %) > LR (62.4 %) > Ensemble model (55.9 %) > RF (53.2 %) > ET (51.3 %) > XGBoost (49.1 %) > DNN (48.1 %) > MLP (47.5 %) > RBF (47.1 %) > KNN (45.8 %) > NB (33.3 %) > SVM (13.7 %) > Bi-LSTM (5.7 %), indicating that DSPA may have the highest Se for predicting sepsis. On the other hand, the SUCRA results for Sp showed that CNN (78.3 %) > CNN + Bi-LSTM (77.6 %) > DSPA (75.1 %) > ET (69 %) > Bi-LSTM (68.5 %) > MLP (51 %) > RBF (50.2 %) > KNN (47.3 %) > RF (47 %) > Ensemble Model (43.4 %) > XGBoost (38.1 %) > SVM (37.3 %) > NB (34.2 %) > DNN (31.1 %) > LR (30.4 %) > Imbalance-XGBoost (21.5 %), suggesting that CNN may have the highest Se for predicting sepsis. Furthermore, the SUCRA results for predictive accuracy showed that DSPA (85.9 %) > CNN + Bi-LSTM (82.6 %) > CNN (81.9 %) > Imbalance-XGBoost (76.8 %) > ET (67.8 %) > RF (51.1 %) > Ensemble model (47.7 %) > XGBoost (44.4 %) > LR (42.7 %) > MLP (38.1 %) > RBF (37.8 %) > KNN (37.3 %) > DNN (35.8 %) > Bi-LSTM (33.3 %) > NB (21.5 %) > SVM (15.3 %) suggesting that DSPA may have the highest accuracy in predicting sepsis probability.

3.3.4. Literature bias

The results for Se and predictive accuracy may have been affected by small sample effects, while those for Sp were not. Based on the shape and symmetry of the funnel plot, we can assess whether the results of the systematic review are affected by literature bias. As shown in Fig. 3, the funnel plot showed that the results of each study for Se, Sp, and predictive accuracy were scattered to varying degrees, rather than concentrated around the mean effect value, and were not completely symmetric, suggesting that there may be some degree of literature bias. In addition, we also saw that the bottom of the funnel plots for Se and predictive accuracy had a scattered distribution. However, the bottom of the funnel plot for Sp did not have a scattered distribution. This suggests that the results for Se and predictive accuracy may have been affected by the small sample effect, while the results for Sp were not affected by the small sample effect.

4. Discussion

A total of seven cohort studies with 16 ML algorithms were included

Table 7
Probability ranking of predictive value among different ML models (SUCRA, %).

ML algorithm	SUCRA (%)		
	Se	Sp	Accuracy
Bi-LSTM	5.7	68.5	33.3
CNN + Bi-LSTM	69.7	77.6	82.6
CNN	67.3	78.3	81.9
DNN	48.1	31.1	35.8
DSPA	77.0	75.1	85.9
Ensemble model	55.9	43.4	47.7
ET	51.3	69.0	67.8
XGBoost	49.1	38.1	44.4
Imbalance-XGBoost	72.9	21.5	76.8
KNN	45.8	47.3	37.3
LR	62.4	30.4	42.7
MLP	47.5	51.0	38.1
NB	33.3	34.2	21.5
RBF	47.1	50.2	37.8
RF	53.2	47.0	51.1
SVM	13.7	37.3	15.3

in this systematic review and NMA, including “Bi-LSTM”, “CNN”, “CNN + Bi-LSTM”, “DNN”, “DSPA”, “LR”, “Ensemble model (LR, RFDT, and XGB)”, “ET”, “XGBoost”, “Imbalance-XGBoost”, “KNN”, “MLP”, “NB”, “RBF”, “RF” and “SVM”. Through consistent modeling, we conducted an NMA and ranked different ML algorithms using Se, Sp, and predictive accuracy as evaluation metrics. We conducted a cross-comparison of different ML algorithms and observed that Bi-LSTM and SVM performed poorly in Se, which was significantly different from other algorithms, and in predictive accuracy, SVM and NB performed poorly, which was significantly different from other algorithms. Finally, we conducted a comprehensive evaluation of different ML algorithms using probability ranking and SUCRA scores and determined that DSPA performed the best in terms of Se and predictive accuracy, while CNN performed best in terms of Sp. We also discovered that traditional ML algorithms (e.g., LR, RF, ET, SVM, etc.) generally perform poorly in predicting sepsis, possibly due to the imbalance and complexity of the dataset, particularly in the department of emergency medicine (Fleuren et al., 2020; M. Li et al., 2022). This systematic review and NMA of the effectiveness of different ML algorithms demonstrated that different ML algorithms had different strengths and weaknesses in the prediction of sepsis and had important implications for future research and practice.

4.1. Current status of research

The management of sepsis is a complex and time-sensitive process that requires the early recognition, diagnosis, and treatment of the underlying infection and its complications (Evans et al., 2021; Levy et al., 2003; Singer et al., 2016). In emergency medicine, dealing with patients with acute and urgent medical problems such as sepsis is a critical area. If not treated promptly, sepsis could lead to organ failure and even death (Lorencio Cardenas et al., 2022; Singer et al., 2016). Therefore, early detection and diagnosis of sepsis are critically important to initiate appropriate interventions and improve patient outcomes (Evans et al., 2021; Guo et al., 2022; Komorowski et al., 2022). However, the diagnosis of sepsis has been a challenge due to the heterogeneity and complexity of clinical manifestations and the lack of specific biomarkers (Evans et al., 2021). Existing research has mainly focused on the changes in the condition and prognosis of severe patients, mainly through the monitoring of traditional clinical and laboratory indicators such as heart rate, blood pressure, body temperature, and white blood cell count (Tran et al., 2020). However, these indicators are often influenced by various factors such as individual differences and environmental factors, making it crucial to seek more accurate and reliable predictors for the management of patient condition and prognosis.

ML techniques have been applied to develop sepsis prediction models based on various data sources, such as electronic health records, vital signs, laboratory tests, and other data (Guo et al., 2022; Zhang et al., 2023). ML is a branch of AI that can learn from data and perform tasks that are difficult for humans, including image recognition, natural language processing, and decision-making. ML algorithms are computer programs that can learn from data and produce outputs such as predictions, classifications, or recommendations (Krishnan et al., 2022; Ngiam & Khor, 2019; Riccardi et al., 2014). Traditional statistical and ML algorithms such as linear regression, RF, and SVM have been used in existing prediction models and methods. However, with the development of AI and DL, more and more studies have begun to adopt advanced DL models such as CNN and recurrent neural networks (RNN) to achieve more accurate and reliable predictions (Shi et al., 2022). In this context, the use of ML algorithms has the potential to improve the diagnosis and management of sepsis by providing more accurate and reliable predictions of sepsis outcomes. However, it is important to note that ML algorithms are not a substitute for traditional clinical judgment and should be used in conjunction with other diagnostic tools. Additionally, the use of ML algorithms should be tailored to the specific needs of each patient and healthcare setting, taking into account individual factors such as patient characteristics and disease severity.

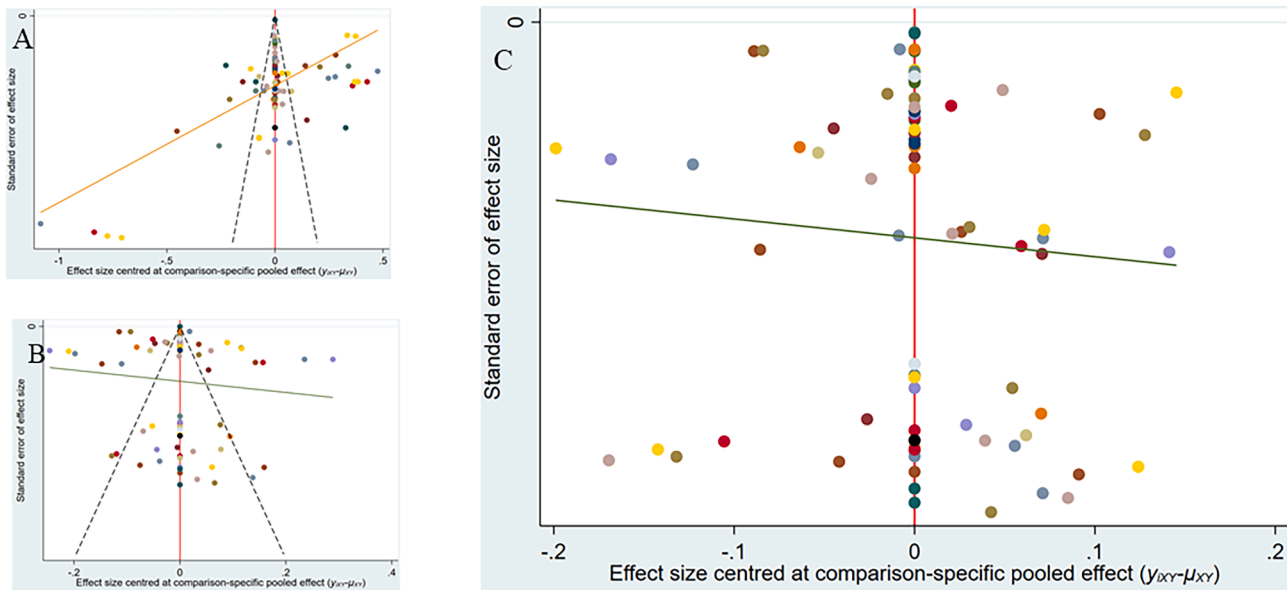


Fig. 3. Funnel plots of Se (A), Sp (B), and predictive accuracy (C).

4.2. Deep learning algorithms

DL algorithms, being a type of AI, can learn from large amounts of data and perform tasks that are difficult for humans, such as image recognition, natural language processing, and decision-making (Asuroglu & Ogul, 2021; Shi et al., 2022). DL algorithms have been shown to play a role in sepsis management, particularly in emergency medicine, by providing data-driven insights, predictions, and recommendations to clinicians, helping them improve the accuracy, efficiency, and outcomes of sepsis care (Fleuren et al., 2020; Jaimes et al., 2005; M. Li et al., 2022). Our results supported the notion that DL algorithms had an advantage in predicting sepsis, as they can automatically extract features from the data without the need for human feature engineering. Our results also indicated that DSPA, with a high SUCRA score for Se (77.0 %) and predictive accuracy (85.9 %), outperformed CNN, which may be due to the fact that DSPA used a recurrent neural network and an attention mechanism to enhance the ML focus on important information, thereby improving its ability to detect early sepsis (Asuroglu & Ogul, 2021; Guo et al., 2022). Our hypothesized that the superior performance of DSPA was attributed to its advanced architecture and design. In summary, DL algorithms hold great promise for improving sepsis care through their ability to extract relevant features from large amounts of data and provide accurate predictions and recommendations to clinicians.

The DSPA algorithm was specifically designed to process the clinical information and physiological parameters of patient to calculate the SOFA score of the patient, which is a clinical scoring system that evaluates the degree of organ dysfunction in six organs (respiratory, cardiovascular, hepatic, coagulation, renal, and neurological) (Asuroglu & Ogul, 2021; Guo et al., 2022; Singer et al., 2016). It has the advantage of automatically extracting features from the data without the need for human feature engineering, as well as improving the ability of ML to focus on important information, thus improving the early detection of sepsis. DSPA and other DL algorithms (such as double coefficient quadratic multivariate fitting function, and data-driven and knowledge-driven ML methods) could be used to monitor the progression and deterioration of sepsis patients, provide early warning of sepsis-related complications such as septic shock or multiple organ failure, and could also provide personalized and adaptive predictions of sepsis risk and mortality, as well as actionable and clinically relevant recommendations for the management of sepsis (Guo et al., 2022; Ma et al., 2023). However, it is important to note that these algorithms are still at an early

stage of development and are in need of further validation and optimisation before they can be used in clinical practice. In addition, the conclusions drawn from this research should be extended to other patient populations and settings, and future lines of research should be carefully discussed to ensure that they are adequately powered and that relevant limitations are addressed. Taking these considerations into account, it may be possible to further improve the accuracy and efficiency of diagnosing and treating sepsis using advanced machine learning algorithms. Ultimately, this approach could lead to improved outcomes for patients with sepsis and a reduction in the burden of this life-threatening condition.

CNN, as a type of DL model, could learn features from complex and high-dimensional data, such as images, signals, or documents (Z. Li et al., 2022). It could be used to analyze various types of data relevant to sepsis diagnosis and prognosis, such as chest X-rays, electrocardiograms, blood tests, or clinical notes. It could also be combined with other types of DL models, such as RNN or attention mechanisms, to capture the temporal dynamics and dependencies of sepsis data and provide accurate and robust predictions of sepsis onset, severity, and outcome, as well as explainable and interpretable results that can help clinicians understand the rationale behind the predictions (Asuroglu & Ogul, 2021; Guo et al., 2022; Z. Li et al., 2022; Rafiei et al., 2021; Shi et al., 2022). In summary, CNN and DSPA are two different sepsis predictive algorithms that could complement each other in sepsis management. CNN provided a comprehensive and explainable prediction of sepsis from different types of data, while DSPA provided a dynamic and personalized prediction of sepsis from a clinical scoring system. Together, they were able to help clinicians improve the diagnosis, prognosis, and treatment of patients with sepsis.

4.3. Imbalance-XGBoost

Imbalance-XGBoost, an ML algorithm based on XGBoost, is designed to handle imbalanced datasets, such as those with a large difference between the number of positive and negative samples (Huang et al., 2021; Shi et al., 2022; Zhang et al., 2023). It uses a cost-sensitive learning approach to assign different weights to different classes according to their proportions in the dataset, in order to improve the classification performance of the minority class. (Liu et al., 2022; Ricciardi et al., 2014; Shi et al., 2022). In addition, our results showed that the Imbalance-XGBoost algorithm, which used heart rate, respiratory rate, oxygen saturation, mean arterial pressure, systolic blood pressure,

diastolic blood pressure, temperature, and blood glucose to construct a 24-dimensional feature vector for each patient, outperformed other traditional ML algorithms in terms of Se and predictive accuracy. This may be because the algorithm used a cost-sensitive learning approach to deal with the imbalance of the data, which improved the classification of minority classes (i.e., septic patients) (Shi et al., 2022; Zhang et al., 2023).

5. Limitations

This NMA had several limitations that should be taken into account in future studies. The sample size of the studies varied widely, from 211 to 22,314, which may be a source of heterogeneity and bias in the results. The methods used in NMA were based on the assumption of consistency between direct and indirect evidence. However, NMA could be affected by the quality and quantity of evidence, choice of model, and confounders. Literature bias and small sample effects were also observed. The NMA did not take into account the differences between the studies in terms of data sources, model training, validation, and implementation.

6. Conclusion

Sepsis is a life-threatening condition that requires early detection and treatment of the infection and its complications (Eskandari et al., 2023). DL algorithms, a type of AI, have the potential to learn from data and perform complex tasks in the diagnosis and management of sepsis (Ibrahim et al., 2020; Liu et al., 2022). The study demonstrated that the DSPA had the highest Se and predictive accuracy for sepsis prediction, indicating that it was the most effective at detecting sepsis cases early and assessing overall sepsis status. The study also demonstrated that CNN had the highest Sp for sepsis prediction, indicating that it was most effective at excluding non-sepsis cases. The study suggested that DSPA and CNN may be the best ML algorithms for sepsis prediction. The study also discussed the imbalance XGBoost algorithm and demonstrated that the Imbalance-XGBoost algorithm outperformed other traditional ML algorithms in terms of Se and predictive accuracy. The study provided a comprehensive and reliable ranking of ML sepsis prediction algorithms, helping clinicians and researchers select the most appropriate for their needs and circumstances. It also demonstrated the potential of DSPA as a tool for predicting sepsis to help improve the diagnosis and management of sepsis and ultimately reduce mortality and morbidity from the condition. Finally, it highlighted the need for further research directions, including exploring alternative ML algorithms that may perform better in sepsis prediction, optimizing existing algorithms to improve their performance, and developing new machine learning methods that can better identify sepsis cases at an early stage. It will also be important to evaluate the practicality and cost-effectiveness of using these algorithms in clinical settings, ensuring that they are suitable for use in resource-limited environments.

7. Ethics approval and consent to participate

As this is a systematic review and NMA, without human involvement, there was no need for ethical approval.

8. Consent for publication

Not applicable.

CRediT authorship contribution statement

Yulei Gao: Conceptualization, Data curation, Formal analysis, Methodology, Software, Writing – original draft preparation, Writing – review & editing, Supervision. **Chaolan Wang:** Data curation, Formal analysis, Software, Writing – original draft preparation, Writing – review

& editing. **Jiaxin Shen:** Data curation, Formal analysis, Methodology. **Ziyi Wang:** Data curation, Formal analysis, Writing – original draft preparation. **Yancun Liu:** Data curation, Writing – review & editing. **Yanfen Chai:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

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Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2023.122982>.

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